

[CH4] Inventing the Computer

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- The outcome of WW2 greatly depended on the technology possessed by the opposing sides
- Massive projects like the Manhattan Project in Los Alamos or the invention of RADAR required large amounts of accurate calculations
- At the start of the war, computing technology was primitive: differential analyzers, punched card machines or teams of human computers. Even the Harvard Mark I hadn't been invented yet
- Computing was not performed for its own sake, but as a means to an end

The Moore School

- Performed ballistics computations for the Aberdeen Proving Ground in Maryland
- The Ballistics Research Laboratory at the proving ground acquired one of the earliest copies of the Vannevar Bush Differential Analyzer
- The BRL and The Moore School of Electrical Engineering both had Differential Analyzers and the BRL used The Moore School's one for its backlog of ballistics calculations
- Even with 2 differential analyzers and hundreds of human computers, the calculations were still a major bottleneck

The Atanasoff-Berry Computer

- Was an obscure project at the Iowa State University, started 1937 by **John Vincent Atanasoff**, and helped by graduate student **Clifford Berry**
- By 1939 they had built a rudimentary electronic computing device
- Over 2 more years they built a complete machine, known as the Atanasoff-Berry Computer (ABC)
- Mauchly was invited by Atanasoff to see the ABC, but he later denied having drawn any ideas from Atanasoff's design

Eckert and Mauchly

- Mauchly established a strong rapport with John Presper Eckert by working on the differential analyzer together, making it work 10 times faster and 10 times more accurately
- Eckert was "*undoubtedly the best electronic engineer at the Moore School*"
- August 1942 Mauchly wrote a memorandum on "*The Use of High Speed Vacuum Tube Devices for Calculating*"
- His memorandum was ignored by both the director of research of the Moore School and the Army Ordnance Department
- Lt. Herman H. Goldstine, a mathematics PhD who joined the BRL in 1942, was responsible for this proposal being taken seriously
- A project was signed between the BRL and the Moore School to build "Project PX", the ENIAC (electronic number integrator and computer)
 - Eckert realised that by reducing the operating voltage of vacuum tubes, he could significantly increase their life expectancy
 - He also noticed that vacuum tubes tended to break down when they were switched on and warmed up. For this reason, the ENIAC tried to never switch off its tube heaters
- The ENIAC had a few major shortcomings:
 - Too little storage
 - Too many tubes
 - Too lengthy reprogramming

- Von Neumann was interested in the project and helped the ENIAC group develop the EDVAC (Electronic Discrete Variable Automatic Computer). This was the first "stored-program computer", on which virtually all computers up to present day have been based

Maurice Wilkes and EDSAC

- Britain was the nation that finished the first two computers
- This was because money was tight at Manchester and Cambridge universities, forcing them to keep things simple. This avoided setbacks that the Americans encountered
- They already had experience: 1940, Alan Turing had specified a mechanical computer at Bletchley Park. 1943 it was followed by an electronic machine, Colossus.
- Turing established the National Physical Laboratory's computing project
- Max Newman established the computer project at Manchester
- The Manchester computer was the first to become operational.
- After the war, Max Newman secured funding to build an EDVAC type computer at Manchester University.
- He was joined by brilliant radio engineer F.C. Williams (Sir Frederic Williams).
- Williams decided that the key problem was memory
- He developed a memory technology based on commercially available cathode-ray tubes
- The team built a tiny computer with no keyboards or printers, only a panel of press-buttons for entering the program bit-by-bit
- This computer was the Manchester Baby and it established the feasibility of the stored-program computer
- Comrie, who was one of the first visitors at the Moore School in 1946, visited Maurice Wilkes at Cambridge University.
- Because of travel restrictions, Wilkes only attended the last 2 weeks of the Moore School Lectures. After that, he visited Harvard and saw the Mark I and the new Differential Analyzer at MIT. He was convinced that these were dinosaurs and that the future lay in electronic stored program computers
- On his way back to the UK, he was already sketching the design of a *"computer of modest dimensions, very much along the lines of the EDVAC proposal"*
- He decided that he wanted the lab staff to become experts in using computers, in programming, not in building them. He wanted to **have** a computer, not to advance computer engineering technology
- He used delay-line storage and created the EDSAC: Electronic Delay Storage Automatic Calculator
- The finished machine had 3000 tubes (only one sixth the tubes of the ENIAC's) and was correspondingly smaller
- Information was inputted and outputted on telegraph tape
- 1949 the EDSAC ran a program to print a table of squares of integers and the world's first practical computer was born, along with the dawn of the computer age